ARDUINO / INDEPENDENT HARDWARE SKETCHES

ESP32 Smart Car Lab

Choose one lab sketch, match the wiring, upload it to an ESP32 and inspect the hardware response.

QUICK START

Open: README sketch inventory

Run locally:

```
Arduino IDE: copy one extensionless source file to a new .ino sketch
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Next: Select your ESP32 board and serial port, compile, then upload.

YOUR FIRST SESSION

01 Install Arduino IDE and Espressif ESP32 board support. Select the specific board you own.

02 Choose one sketch from the README inventory; the experiments are independent and should not be combined.

03 The Bluetooth car advertises ESP32_CAR. Motor pins are 13, 12, 14 and 27; the LCD address is 0x27.

04 Verify your motor driver, supply, shared ground, LCD and pin connections against the source and hardware datasheets before power-up.

05 Install libraries imported by the selected sketch, such as LiquidCrystal_I2C. Compile and upload one experiment.

06 Connect a Bluetooth serial controller and send F, B, L, R or S for movement.

DATA AND PRIVACY

Use an appropriate motor supply and verify the driver and wiring before powering hardware. Incorrect connections can damage components.

LIMITS TO KEEP IN MIND

Hardware was not revalidated. The Range_detector_using_rotator file is not a servo scanner and its PWM command experiment needs review. No Android APK is included.

READ NEXT

README sketch inventory; inspect each source before wiring

Steps were checked against the repository README and source. Review local configuration, privacy and hardware before using real data or devices.